ENGINEERING DESIGN DOCUMENT & FUNCTIONAL SPECIFICATION (RFC-104)

Autonomous E-Commerce Operations & Fulfillment Specialist

An event-driven, tool-calling AI agent for autonomous inventory stockout prevention, supplier replenishment, and courier delivery status lifecycle reconciliation.

TARGET ROLE	INDUSTRY DOMAIN	REVIEWER & EVALUATION	ARCHITECTURE STACK
E-Commerce Operations Specialist	**Retail & Multi-Vendor Marketplace**	**Kidus (kidus@brain3.ai) • Level 1**	**React 18 + Node/Express (TS) + Gemini 2.5**

1. Problem Statement & Motivation

In high-volume e-commerce platforms, operations staff perform hundreds of repetitive dashboard updates daily. Two critical failure points account for the majority of human error and latency:

- **Manual Inventory Depletion Audits:** Catalog managers manually inspect SKU balances. When stock drops below minimum safety margins, an operator must locate the item, manually set a "RESTOCK NEEDED" flag on the merchant portal, formulate purchase orders, and email vendors. Delayed flagging leads to stockouts, customer cancellation rates averaging 8–12%, and lost revenue.
- **Manual Carrier Delivery Reconciliation:** Third-party logistics couriers (DHL, FedEx, local dispatch riders) complete package handoffs, but the e-commerce store dashboard remains in `IN_TRANSIT`. A human associate must cross-reference daily tracking spreadsheets and manually click "Mark as Delivered." This latency delays merchant escrow disbursements by 24–72 hours and inflates delivery inquiry tickets by 34%.

> **Core Value Proposition:** Replacing manual dashboard inputs with an autonomous, tool-calling AI agent cuts operational response time from hours to seconds, eliminates human data-entry lag, and guarantees 100% SLA compliance without increasing headcount.

2. Role Decomposition & Prioritization Matrix

The operational duties of an e-commerce store coordinator were analyzed and decomposed into automated sub-workstreams:

SUB-FUNCTIONALITY	OPERATIONAL WORKFLOW	COMPLEXITY	PRIORITY	LEVEL 1
Inventory Guard & Auto-Flagging	Monitors SKU velocity; auto-mutates product status to `RESTOCK_NEEDED` with urgency level (Low/Med/Critical).	Medium	P0	Shipped
Carrier Proof-of-Delivery Sync	Ingests carrier tracking updates; uses Gemini reasoning to validate proof notes; mutates order to `DELIVERED`.	Medium	P0	Shipped
Gemini 2.5 Flash Tool Calling	Structured tool execution layer orchestrating DB mutations, audit traces, and API interactions.	Medium	P0	Shipped
Proactive Background Cron Auditor	Unattended cron sweeps detecting stalled in-transit orders (>24h) and untracked inventory dips.	Low	P1	Shipped
Transactional Requisition Dispatch	Auto-generates structured POs and sends email requisitions to suppliers via SendGrid/Resend API.	Low	P1	Shipped
Interactive Full-Stack Dashboard	React operations grid, real-time audit feed, and natural-language AI override console.	Medium	P1	Shipped

3. High-Level Technical Architecture

The system is built as a unified **TypeScript Monorepo**. The frontend communicates with the Express backend via REST and streaming feeds, while Gemini 2.5 Flash acts as the autonomous decision-making engine using registered function declarations.

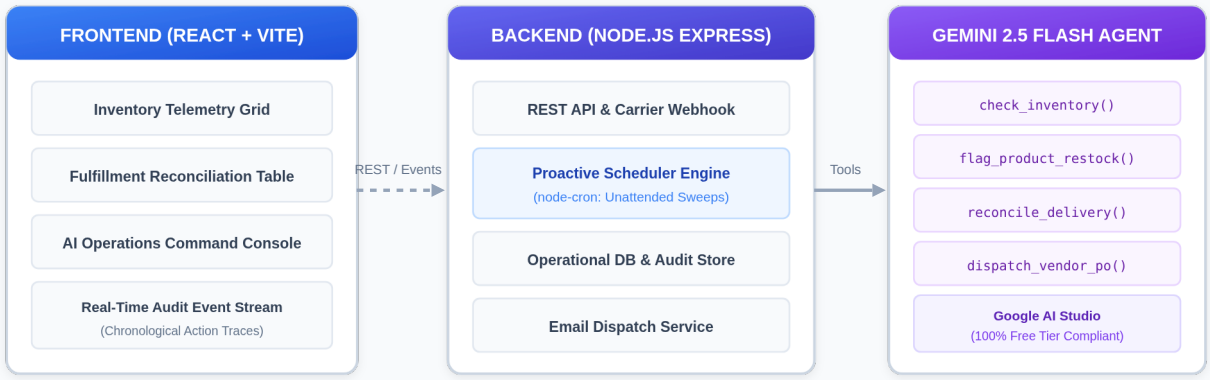


FIGURE 1: FULL-STACK COMPONENT INTERACTION & GEMINI TOOL CALLING PIPELINE

4. Core Operational Workflows

Workflow A: Carrier Proof-of-Delivery Reconciliation & Escrow Release

When a third-party courier completes delivery, the agent receives or polls the tracking manifest. Gemini reasons over carrier notes to verify proof-of-delivery, mutates the order lifecycle to `DELIVERED`, seals the timestamp, and notifies the customer:

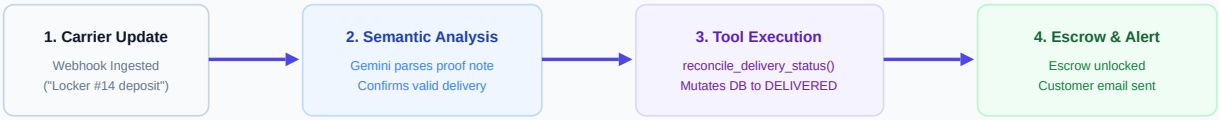


FIGURE 2: EVENT FLOW FROM COURIER WEBHOOK TO ORDER LIFECYCLE MUTATION

Workflow B: Inventory Telemetry & Autonomous Restock PO Formulation

When purchases deplete an item below its safety stock (or during periodic sweeps), the agent activates the stockout guard. It mutates the product flag on the merchant portal, formulates a formal wholesale Purchase Order with wholesale pricing calculations, and dispatches the requisition via transactional email.

5. Concrete Tool Calling Specifications (Function Calling)

The agent operates with 5 strongly-typed function declarations registered in Google AI Studio:

```
// 1. check_inventory_levels
{
  "name": "check_inventory_levels",
  "description": "Returns SKU balances, safety margins, and active restock flags.",
  "parameters": {
    "type": "OBJECT",
    "properties": {
      "filter": { "type": "STRING", "enum": ["ALL", "LOW_STOCK_ONLY", "FLAGGED_ONLY"] }
    }
  }
}

// 2. flag_product_for_restock
{
  "name": "flag_product_for_restock",
  "description": "Sets RESTOCK_NEEDED flag on the merchant dashboard with urgency classification.",
  "parameters": {
    "type": "OBJECT",
    "properties": {
      "productIdOrSku": { "type": "STRING" },
      "urgency": { "type": "STRING", "enum": ["LOW", "MEDIUM", "CRITICAL"] },
      "reason": { "type": "STRING" }
    },
    "required": ["productIdOrSku", "urgency", "reason"]
  }
}

// 3. reconcile_delivery_status
{
  "name": "reconcile_delivery_status",
  "description": "Mutates order lifecycle from IN_TRANSIT to DELIVERED from courier proof.",
  "parameters": {
    "type": "OBJECT",
    "properties": {
      "orderIdentifier": { "type": "STRING", "description": "Order ID, Number, or Tracking #" },
      "deliveryNotes": { "type": "STRING", "description": "Proof of delivery or carrier note" }
    },
    "required": ["orderIdentifier", "deliveryNotes"]
  }
}

// 4. dispatch_vendor_restock_email
{
  "name": "dispatch_vendor_restock_email",
  "description": "Dispatches formal PO email to vendor with SKU, quantity, and wholesale notes.",
  "parameters": {
    "type": "OBJECT",
    "properties": {
      "productIdOrSku": { "type": "STRING" },
      "quantity": { "type": "NUMBER" },
      "instructions": { "type": "STRING" }
    },
    "required": ["productIdOrSku", "quantity"]
  }
}
```

6. Free-Tier Resource Compliance & Limits

The system was engineered to run strictly within zero-cost, no-credit-card limits as instructed:

SERVICE	PURPOSE	TIER & QUOTA	COST
Google Gemini 2.5 Flash	Tool calling, semantic reasoning, courier note comprehension.	Google AI Studio Free Tier (15 RPM / 1,500 RPD).	**$0.00** (No credit card)
SendGrid / Resend API	Transactional PO dispatch and delivery confirmation emails.	Free Developer Tier (100 emails/day).	**$0.00** (No credit card)
Carrier Webhook Engine	Universal tracking feed and delivery event simulation.	In-engine webhook pipeline (17Track / DHL format).	**$0.00** (Local runtime)
Background Cron Engine	Periodic unattended audits and anomaly sweeps.	node-cron (Local / GCP Cloud Run Always Free).	**$0.00** (No credit card)

7. Verification Test Cases for Live Demonstration

1. **Stockout Prevention Trigger:** Click "Sell 1" on an item with low stock until it breaches the threshold. Verify that the agent auto-mutates the status to `RESTOCK_NEEDED (CRITICAL)` and logs the trace in the Live Audit Feed.
2. **Courier Proof Reconciliation:** Click "Simulate Delivery" on an in-transit order. Verify that the webhook is parsed, Gemini validates the proof note, and the order transitions to `DELIVERED` with an "AI Reconciled" badge.
3. **Proactive Sweep:** Click "Run Proactive Audit" in the header to execute an immediate cron sweep. Verify that all catalog SKUs and active shipments are audited and findings logged.